



ATLAS CAMERA · SHOWCASE DEEP-DIVE 09 / 11 [← ALL WORKFLOWS](#)

ORGANIC RELIEF + RETOPO — JUNGLE RUINS

Canopy doctrine in, sculpt-ready quads out: grid 512, loose edges, hole-fill, quad remesh, textured OBJ+GLB.

FILE `atlas_organic_relief_jungleruins_workflow.json` VERIFIED 2026-07-16

THE STORY

Dense canopy depth is genuinely noisy at small scale. The relief doctrine for organics: raise the grid (512 here) so each quad spans less real-world area, and LOOSEN `depth_edge_rel` to 1.0 — the default 0.5 reads canopy noise as depth discontinuities and shreds the mesh into holes. An `AtlasBoundedBand` clips the foreground's own extrusion so monocular 'banana' error can't run the ruins off into the distance.

The export is where this workflow earns its place: export-only hole-fill caps small interior tears (the live projection mesh keeps its deliberate silhouette tears — that doctrine is untouched), then QUAD retopology remeshes down to ~4,000 target vertices with projective UVs regenerated by inverting the camera bake — so the retopologized mesh still imports TEXTURED into Maya, ZBrush, or Blender. `build_scene.py` reproduces the whole setup in Blender's Script editor.

IN THE VIEWPORT — PROJECTED VS. THE GREY MESH



SOURCE PLATE

jungleruins_32bit_acescg.exr · ACEScg float — the single still photograph everything below is reconstructed from.



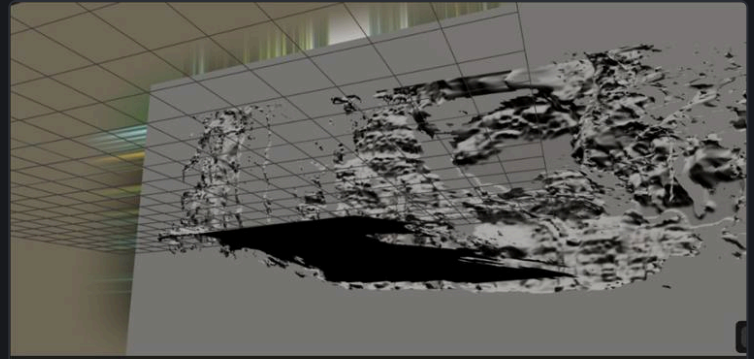
PROJECT ON

The photo reassembled on the recovered camera — texels assigned by ray, perfect from this view.



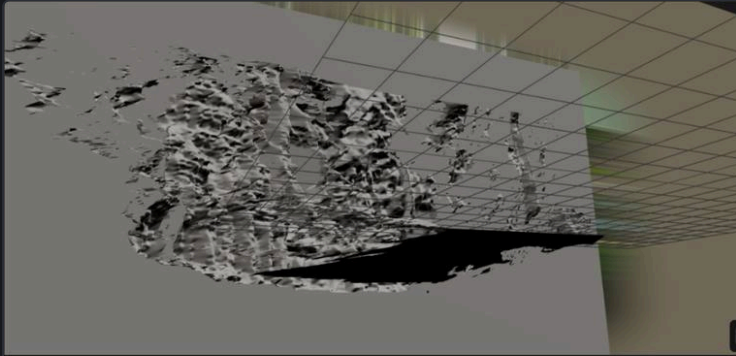
PROJECT OFF

The same geometry as a grey mesh, recovered view — what the depth-derived surfaces actually are.



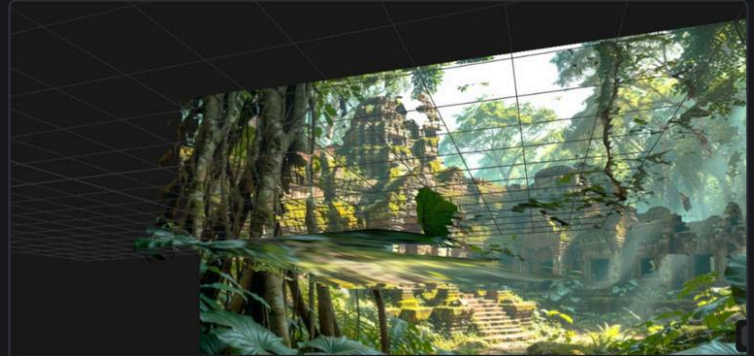
GREY · ORBIT LEFT

Orbited off-axis: the reconstructed cone. Coverage ends where the camera never saw.



GREY · ORBIT RIGHT

The opposite angle — silhouette tears read as deliberate holes, not errors.



BAND BOX

🔍 overlay: the translucent box's back face is the metric cutoff plane the layer is clipped at.

NODES ON STAGE

OCIORead

AtlasDepthMap

AtlasDeriveReliefMesh

AtlasBoundedBand

AtlasExportReliefMesh

AtlasExportBlender

RETUNE ON YOUR PLATE

- **relief_grid / depth_edge_rel — the organic dial pair.** Denser canopy wants a finer grid AND a looser edge threshold. Watch the hole fraction, not the vert count.
- **retopo_method + retopo_target_vertex_count.** quad (pyinstantmeshes) for sculpting; decimate for lightweight look-dev; smooth to relax without changing topology.
- **fill_interior_holes + the depth window.** fill_depth_near/far_m scopes which tears get capped — transcribe AtlasBoundedBand's cutoff_m to fill only inside the subject.
- **Preview before the DCC round-trip.** Wire the export's preview_solve into a viewport — it carries the mesh ACTUALLY written, fills and all.

RUN ✓ Quad retopo landed at 4,493 verts (target 4,000); OBJ and GLB import textured — projective UVs regenerated and pinned to 1e-5; build_scene.py ready for Blender.

REQUIREMENTS

OCIO/EXR env · pyinstantmeshes + fast-simplification (retopo) · [neural]

Atlas Camera v0.6.0 beta · [examples/showcase](#)

[Master field guide](#) →



ATLAS CAMERA · RECOVER THE CAMERA FROM A SINGLE STILL